

Registro de Calibración del Software (según IEC 61083-2)

1. Información de la Validación

Software Probado: Analizador de Impulsos atmosféricos tipo rayo (1.2/50 μ s). (v1.0.0) - 2026-03-24

Algoritmos soportados: Full Lightning Impulse (LI), Chopped Lightning Impulse (LIC)

Parámetros validados: Valor Pico (Ut), Tiempo de Frente (T1), Tiempo de Cola (T2), Sobrepasamiento (OS)

Generador de Datos de prueba (TDG): IEC 61083-2 TDG Version 2.1.0

Resolución TDG: 16 bits | Tasa de Muestreo: 100 MSa/s

2. Tabla de Resultados

2.1 Full Lightning Impulses - LI

Onda	Ut [kV]			T1 [μ s]			T2/Tc [μ s]			OS [%]		
	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv
LI-A1	1049.60	1049.59	-0.001%	0.840	0.840	-0.029%	60.16	60.15	-0.009%	0.00	-0.00	-0.000%
LI-A2	1037.60	1037.61	0.001%	1.693	1.692	-0.059%	47.48	47.48	0.001%	5.10	5.14	0.041%
LI-A3	1000.20	1000.15	-0.005%	1.117	1.116	-0.115%	48.15	48.16	0.011%	4.60	4.57	-0.029%
LI-A4	856.01	856.01	0.000%	0.841	0.840	-0.134%	47.80	47.80	0.007%	7.90	7.89	-0.011%
LI-A5	71.97	71.97	0.000%	1.711	1.714	0.165%	47.71	47.71	-0.004%	7.70	7.76	0.060%
LI-A6	100.17	100.17	0.000%	1.762	1.766	0.243%	41.58	41.58	-0.000%	17.70	17.75	0.047%
LI-A7	104.35	104.35	-0.001%	2.122	2.127	0.225%	38.36	38.37	0.018%	20.10	20.18	0.082%
LI-A8	96.01	96.01	0.002%	1.503	1.506	0.212%	44.92	44.93	0.013%	14.80	14.77	-0.034%
LI-A9	55.93	55.93	-0.004%	1.215	1.216	0.057%	55.74	55.74	0.001%	4.00	4.03	0.032%
LI-A10	81.93	81.93	0.001%	0.924	0.926	0.173%	42.66	42.66	0.002%	12.00	12.03	0.029%
LI-A11	86.60	86.60	0.001%	0.578	0.577	-0.196%	56.37	56.37	-0.005%	4.10	4.07	-0.030%
LI-A12	85.58	85.58	0.000%	0.587	0.588	0.149%	57.36	57.36	0.001%	2.30	2.27	-0.030%
LI-M1	952.09	952.06	-0.003%	1.123	1.124	0.093%	85.60	85.61	0.013%	2.10	2.08	-0.018%
LI-M2	-1041.70	-1041.59	-0.011%	3.356	3.354	-0.053%	61.25	61.25	-0.000%	9.20	9.19	-0.012%
LI-M3	-1026.50	-1026.42	-0.007%	2.150	2.148	-0.079%	41.75	41.75	0.003%	9.20	9.18	-0.022%
LI-M4	-267.14	-267.13	-0.003%	0.987	0.987	0.041%	56.22	56.22	-0.004%	4.80	4.84	0.036%
LI-M5	-55.00	-54.99	-0.029%	2.746	2.750	0.161%	42.11	42.11	-0.002%	18.70	18.70	-0.004%
LI-M6	-166.87	-166.87	0.000%	1.356	1.356	-0.027%	54.74	54.74	-0.002%	3.80	3.83	0.030%
LI-M7	-1272.30	-1272.05	-0.019%	1.482	1.485	0.203%	50.03	50.04	0.021%	11.20	11.19	-0.010%
LI-M8	-99.73	-99.73	0.001%	1.515	1.515	0.013%	49.36	49.36	-0.001%	-0.50	-0.55	-0.054%
LI-M9	-100.04	-100.03	-0.009%	0.828	0.828	-0.005%	46.65	46.65	0.008%	1.40	1.38	-0.021%
LI-M10	100.26	100.26	0.002%	1.666	1.667	0.049%	60.85	60.85	0.005%	0.00	-0.00	-0.004%
LI-M11	299.32	299.32	-0.001%	1.661	1.661	-0.009%	60.95	60.94	-0.015%	-0.50	-0.46	0.044%
LI-M12	-4.32	-4.32	-0.007%	1.292	1.293	0.100%	52.27	52.27	-0.008%	-1.80	-1.78	0.021%
LI-M13	39.46	39.46	-0.003%	1.537	1.538	0.080%	46.94	46.94	-0.004%	1.80	1.77	-0.031%
LI-M14	48.55	48.55	-0.007%	0.933	0.934	0.079%	37.48	37.47	-0.015%	4.30	4.26	-0.043%
LI-M15	497.97	497.98	0.002%	1.017	1.017	0.021%	59.19	59.19	-0.002%	-0.10	-0.07	0.028%
LI-M16	369.21	369.20	-0.004%	0.920	0.920	-0.028%	47.53	47.53	-0.003%	0.80	0.83	0.034%
LI-M17	-99.35	-99.35	0.002%	1.775	1.774	-0.054%	53.31	53.31	0.006%	1.30	1.33	0.027%

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2.2 Chopped Lightning Impulses (LIC)

Onda	Ut [kV]			T1 [μs]			T2/Tc [μs]			OS [%]		
	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv
LIC-M4	0.15	0.15	-0.027%	1.305	1.305	-0.001%	6.00	6.01	0.122%	-0.20	0.60	0.797%
LIC-M5	-389.90	-389.85	-0.013%	0.857	0.868	1.275%	9.24	9.27	0.296%	6.80	6.90	0.105%

3. Estimación de Incertidumbre (según Anexo B.3.3)

Basado en las máximas desviaciones encontradas:

3.1 Full Lightning Impulses - LI

Parámetro	u_B71 (Software) [%]	u_B72 (Referencia) [%]	u_B7 (Combinada) [%]
Valor Pico (Ut)	0.0169	0.0150	0.0226
Tiempo de Frente (T1)	0.1404	0.2500	0.2867
Tiempo de Cola (T2/Tc)	0.0120	0.0300	0.0323
Sobrepasamiento (OS)	0.0473	0.0400	0.0619

3.2 Chopped Lightning Impulses (LIC)

Parámetro	u_B71 (Software) [%]	u_B72 (Referencia) [%]	u_B7 (Combinada) [%]
Valor Pico (Ut)	0.0158	0.0250	0.0296
Tiempo de Frente (T1)	0.7363	0.4500	0.8629
Tiempo de Cola (T2/Tc)	0.1707	0.1500	0.2272
Sobrepasamiento (OS)	0.4600	0.0250	0.4607